Power Query in Excel

Interview Questions and Answers

Importing separate CSV files • Merge Queries • Join types • Data quality

These questions are designed around a practical exercise that merges Students.csv and Courses.csv using SID as the common key. Use the answers for interview preparation and classroom discussion.

Section 1: Basic Interview Questions

1. What is Power Query in Excel?

Answer: Power Query is an Excel tool used to import, clean, transform, combine, and prepare data from CSV files, workbooks, databases, folders, and other sources.

2. What is the difference between Merge Queries and Append Queries?

Answer: Merge combines tables horizontally by matching common columns. Append combines tables vertically by stacking rows.

3. Can we merge two separate CSV files in Power Query?

Answer: Yes. Import the first CSV using Data → Get Data → From File → From Text/CSV. In Power Query Editor, use Home → New Source to import the second CSV, then choose Merge Queries.

4. What is a join key?

Answer: A join key is the column used to match records between two tables. In this example, SID connects the Students and Courses tables.

5. What is the difference between a primary key and a foreign key?

Answer: A primary key uniquely identifies a record in a table. A foreign key refers to a key in another table. SID may be a primary key in a student master table and a foreign key in a course registration table.

Section 2: Join Types

6. What is a Left Outer Join?

Answer: It retains all rows from the first table and adds matching rows from the second. If there is no match, columns from the second table contain null values.

7. What is a Right Outer Join?

Answer: It retains all rows from the second table and adds matching records from the first. Unmatched records from the second table remain in the output.

8. What is an Inner Join?

Answer: It returns only records that have matching key values in both tables.

9. What is a Full Outer Join?

Answer: It retains all records from both tables. Matching records are combined; unmatched records have nulls in columns from the other table.

10. What is the difference between Left Anti Join and Right Anti Join?

Answer: Left Anti returns rows from the first table without a match in the second. Right Anti returns rows from the second table without a match in the first.

Section 3: Practical and Scenario-Based Questions

11. If you perform a Left Outer Join, what happens to SID 4?

Answer: Ramesh (SID 4) remains in the output, but Course is null because there is no matching SID 4 in Courses.csv.

12. Why might Mayank appear twice after a merge?

Answer: SID 1 appears twice in Courses.csv—once for Python and once for Excel. The merge produces one output row for each matching course.

13. Which join identifies students without course registrations?

Answer: Use Left Anti Join, with Students as the first table and Courses as the second.

14. Which join identifies course records without a corresponding student?

Answer: Use Right Anti Join, with Students as the first table and Courses as the second. SID 5 (Tableau) would be returned.

15. Why should you check duplicate IDs before merging?

Answer: Duplicate keys can produce multiple output rows for one record. This may be expected in a one-to-many relationship, but unexpected duplicates can inflate totals.

Section 4: Data Quality and Troubleshooting

16. What should you check if a merge returns unexpected results?

Answer: Verify the selected join columns, data types, extra spaces, inconsistent key formats, duplicate values, null keys, and selected join kind.

17. What happens if one SID column is Text and the other is Whole Number?

Answer: They have incompatible data types for reliable matching. Convert both columns to compatible types before merging.

18. How do you expand a merged table?

Answer: Click the expand icon in the new nested-table column, select the required fields (such as Course), and click OK.

19. How do you load the merged result into Excel?

Answer: In Power Query Editor, select Home → Close & Load. The output is loaded into an Excel worksheet or table.

20. How do you refresh the merged result when the source data changes?

Answer: Use Data → Refresh All in Excel, provided the source files remain accessible at their configured paths.

Section 5: Advanced Interview Questions

21. What is a one-to-many relationship?

Answer: One record in one table can match multiple records in another table.

22. Can merging tables increase the number of rows?

Answer: Yes. Multiple matches for a key can generate multiple output rows.

23. What is the difference between null and zero?

Answer: Null represents a missing or unknown value; zero is a numeric value.

24. How would you combine monthly sales CSV files with the same columns?

Answer: Use Append Queries, or combine files from a folder in Power Query.

25. Why should you validate row counts after merging?

Answer: To check that the output aligns with the join type and the expected relationships between keys.

Join Types at a Glance

Join type	Rows retained	Example use
Left Outer	All rows from Table 1	Keep all students and matching courses
Right Outer	All rows from Table 2	Keep every course record
Inner	Only matching rows	Find records present in both tables
Full Outer	All rows from both tables	Reconcile two lists
Left Anti	Unmatched rows from Table 1	Find students without course records
Right Anti	Unmatched rows from Table 2	Find course records without student records

Practical Interview Exercise

Ask students to complete these tasks in Excel Power Query:

1. Import Students.csv and Courses.csv as separate queries.
2. Perform a Left Outer Join on SID.
3. Repeat the merge using Inner Join and Full Outer Join.
4. Use Left Anti Join to find students without matching course records.
5. Use Right Anti Join to find course records without matching students.
6. Explain why duplicate keys can increase the output row count.
7. Expand the Course column and load the result into Excel.

Interview Preparation Tips

- Explain each join in your own words and describe which rows are retained.
- Be prepared to demonstrate a merge using two separate CSV files.
- Check data types, duplicate keys, null values, and row counts after merging.
- Explain unexpected results using the relationships between keys in both tables.